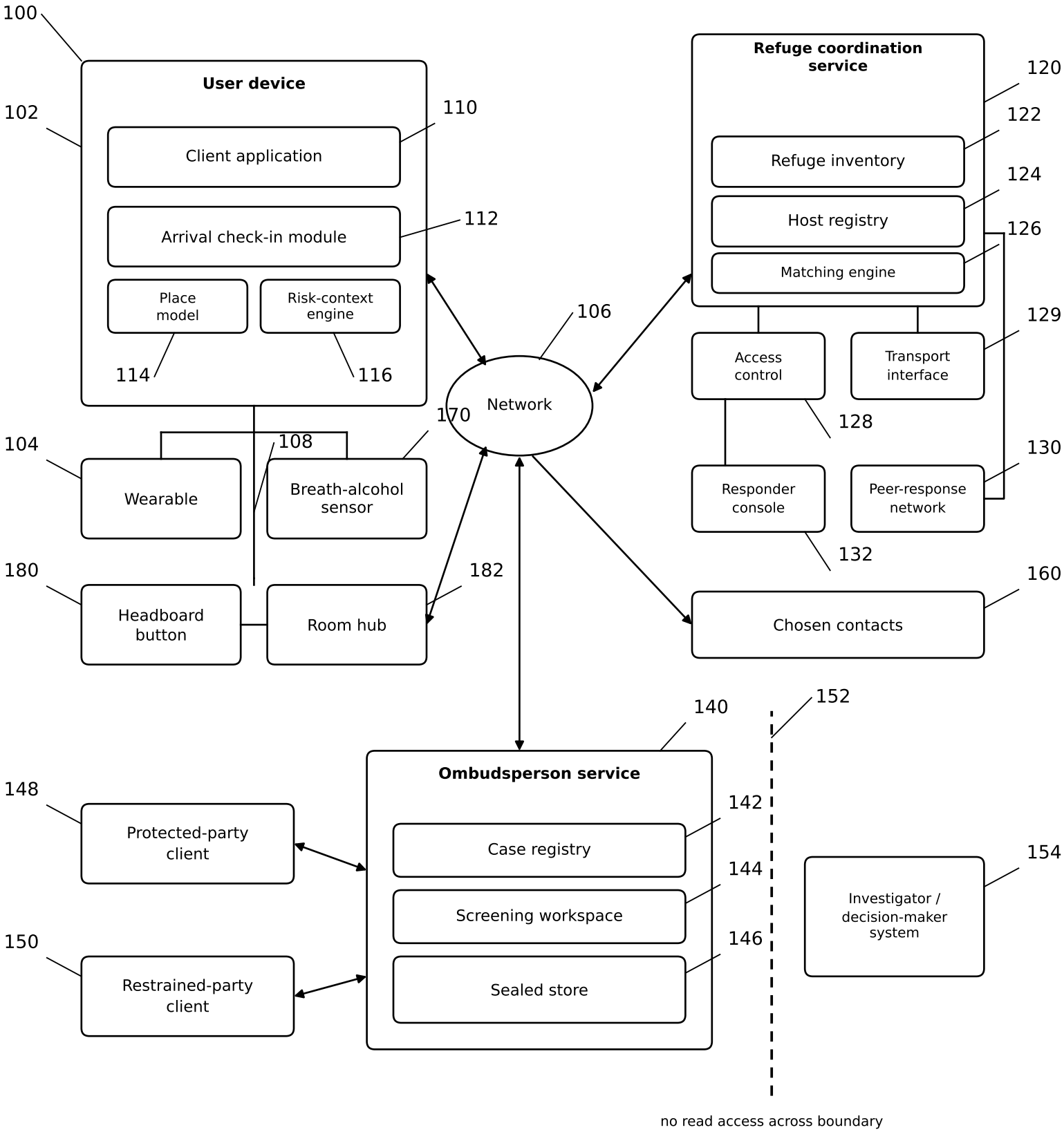

100
User device
102
110
Client application
Arrival check-in module
112
Place model
Risk-context engine
114
116
106
Network
170
104
108
Wearable
Breath-alcohol sensor
180
182
Headboard button
Room hub
Refuge coordination service
120
122
Refuge inventory
124
Host registry
126
Matching engine
129
Access control
Transport interface
128
130
Responder console
Peer-response network
132
160
Chosen contacts
140
152
Ombudsperson service
148
142
Protected-party client
Case registry
144
Screening workspace
154
Investigator / decision-maker system
150
146
Restrained-party client
Sealed store
no read access across boundary

FIG. 1

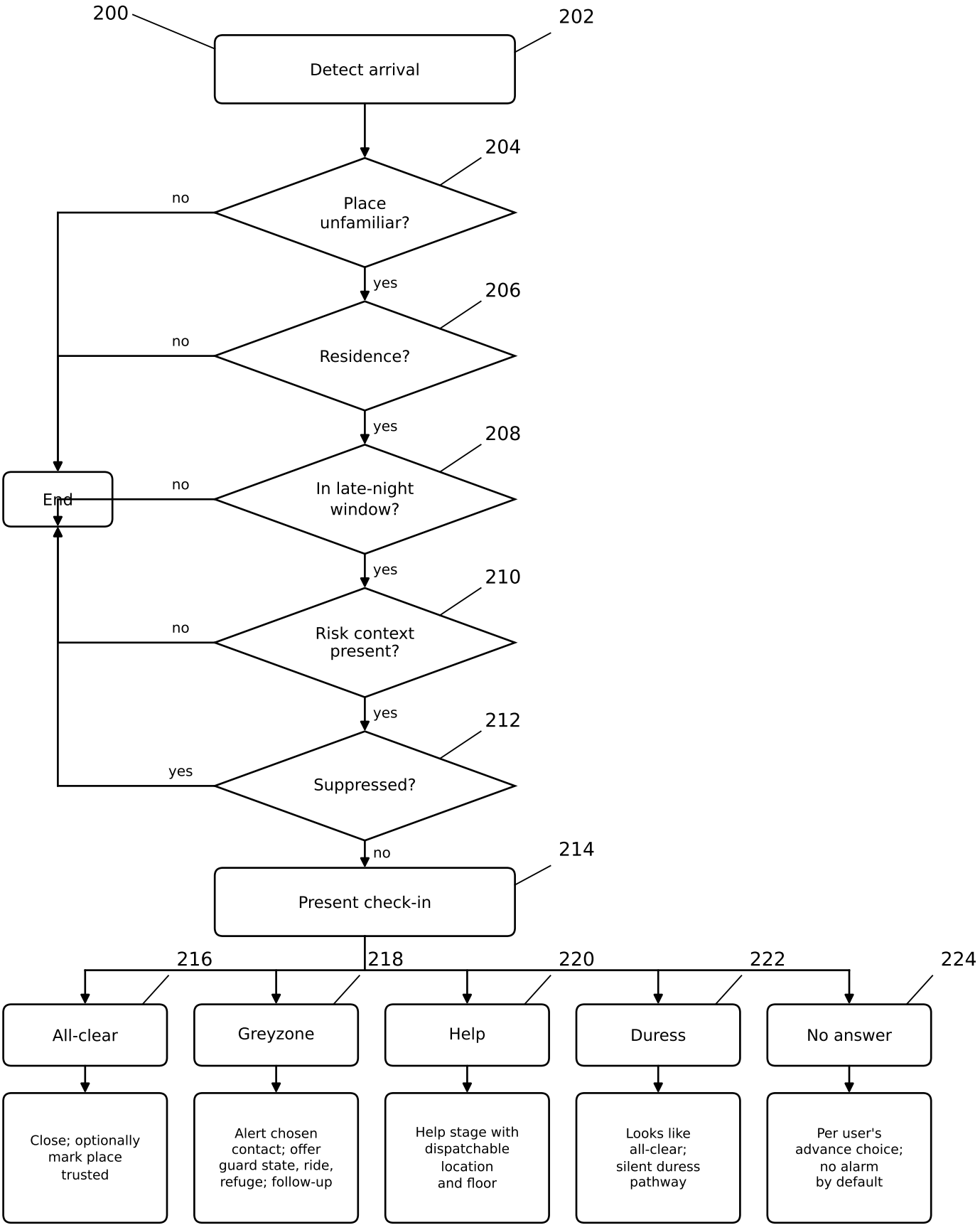

200
202
Detect arrival
204
Place unfamiliar?
no
yes
206
Residence?
no
yes
208
In late-night window?
no
End
yes
210
Risk context present?
no
yes
212
Suppressed?
yes
no
214
Present check-in
216
218
220
222
224
All-clear
Greyzone
Help
Duress
No answer
Close; optionally mark place trusted
Alert chosen contact; offer guard state, ride, refuge; follow-up
Help stage with dispatchable location and floor
Looks like all-clear; silent duress pathway
Per user's advance choice; no alarm by default

FIG. 2

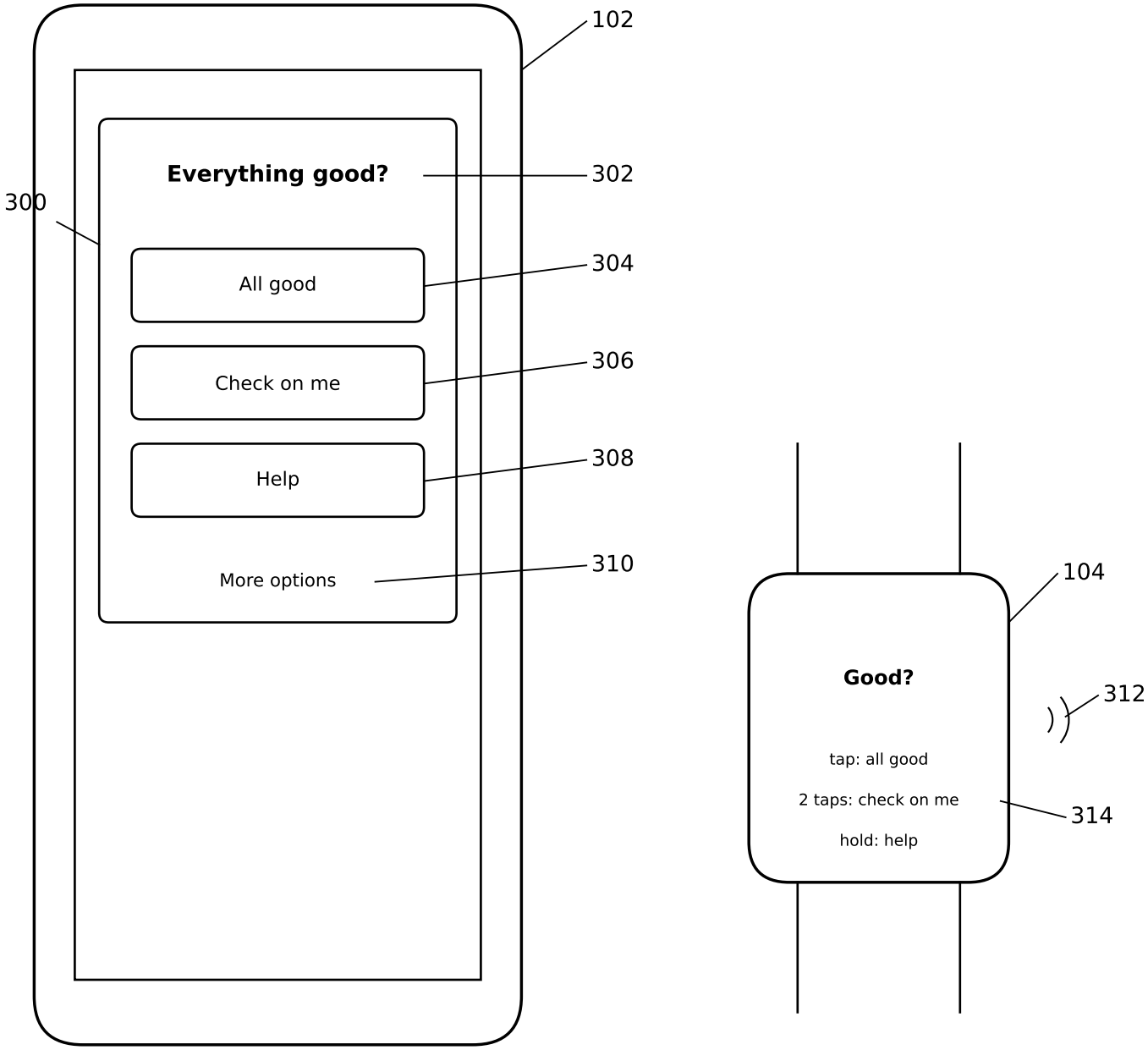

102
Everything good?
302
300
All good
304
Check on me
306
Help
308
More options
310
104
Good?
312
tap: all good
2 taps: check on me
hold: help
314

FIG. 3

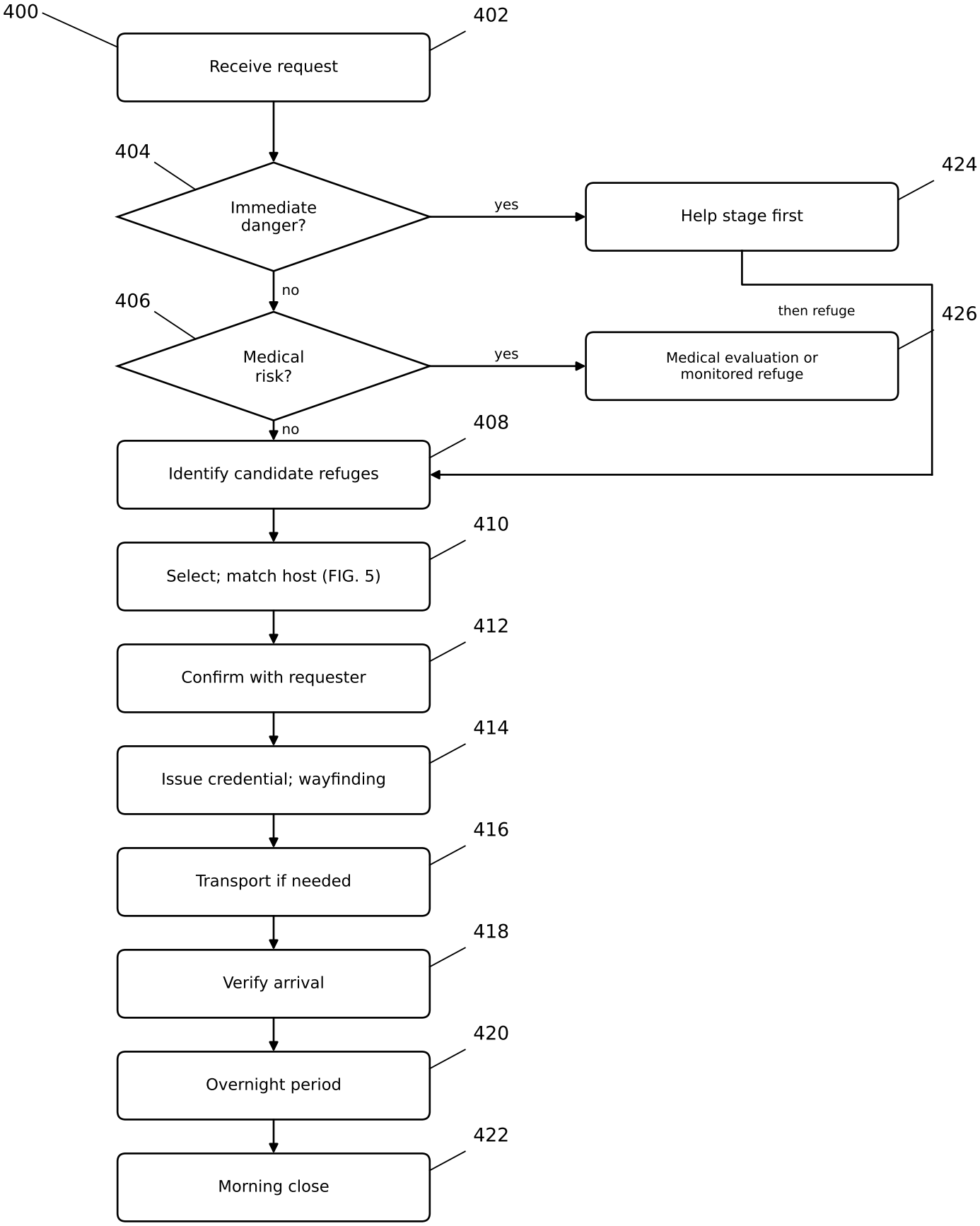

400
402
Receive request
404
Immediate danger?
yes
424
Help stage first
no
406
then refuge
426
Medical risk?
yes
Medical evaluation or monitored refuge
no
408
Identify candidate refuges
410
Select; match host (FIG. 5)
412
Confirm with requester
414
Issue credential; wayfinding
416
Transport if needed
418
Verify arrival
420
Overnight period
422
Morning close

FIG. 4

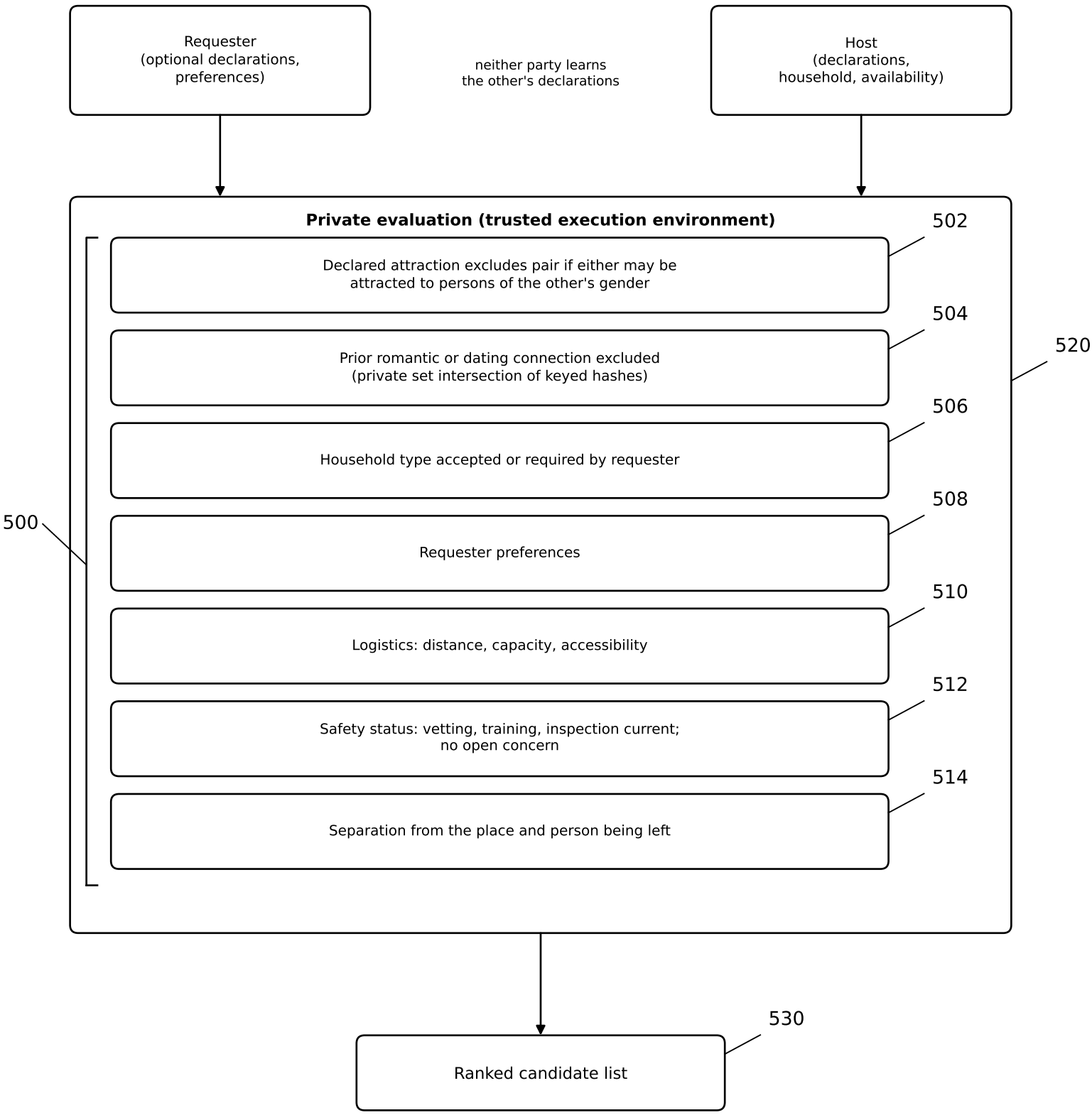

Requester
(optional declarations,
preferences)
neither party learns
the other's declarations
Host
(declarations,
household, availability)
Private evaluation (trusted execution environment)
Declared attraction excludes pair if either may be
attracted to persons of the other's gender
Prior romantic or dating connection excluded
(private set intersection of keyed hashes)
Household type accepted or required by requester
Requester preferences
Logistics: distance, capacity, accessibility
Safety status: vetting, training, inspection current;
no open concern
Separation from the place and person being left
Ranked candidate list
500
502
504
506
508
510
512
514
520
530

FIG. 5

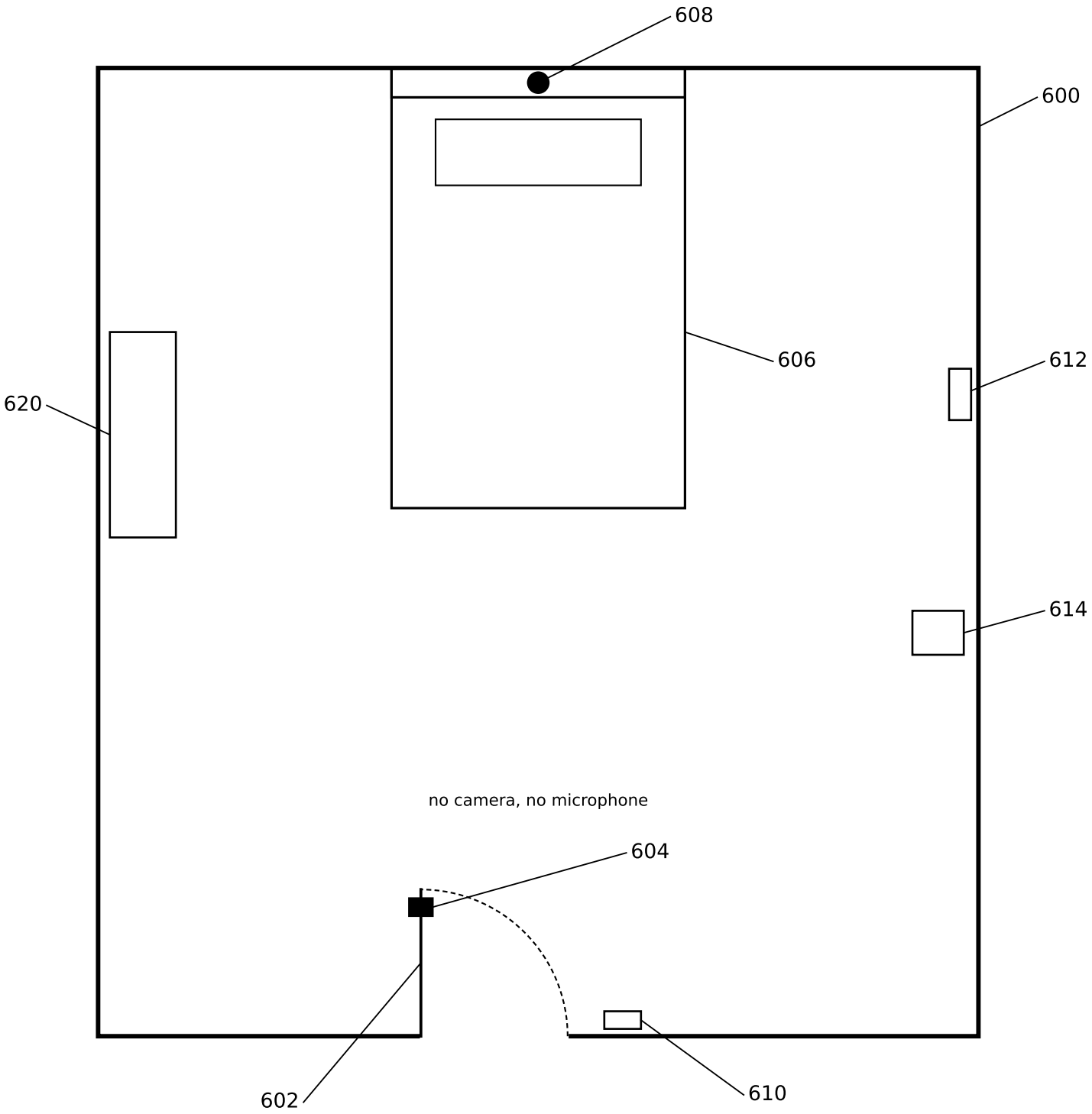


FIG. 6

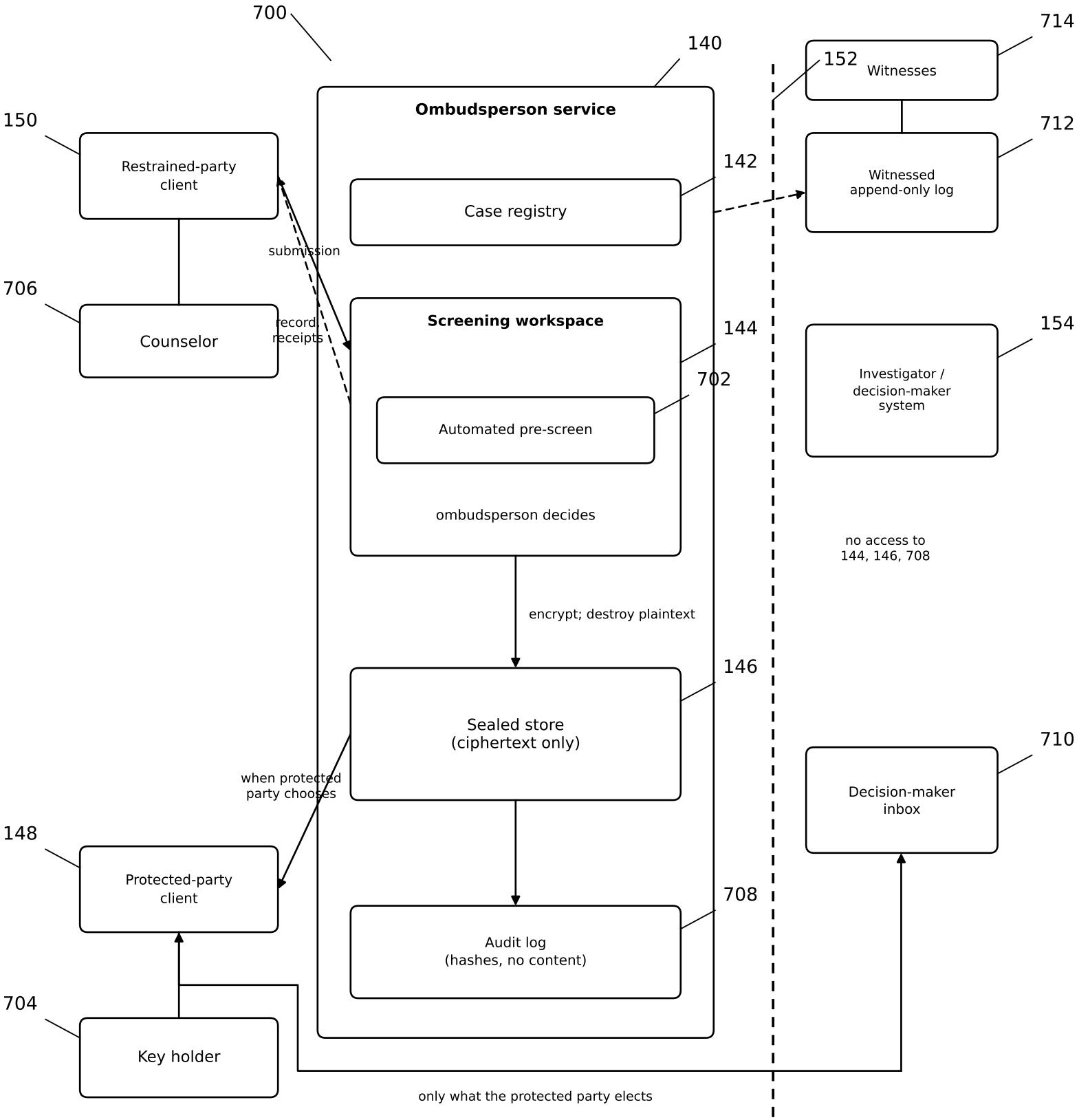

700
140
Ombudsperson service
150
Restrained-party client
706
Counselor
submission
record receipts
142
Case registry
144
Screening workspace
702
Automated pre-screen
ombudsperson decides
encrypt; destroy plaintext
146
Sealed store (ciphertext only)
when protected party chooses
148
Protected-party client
704
Key holder
708
Audit log (hashes, no content)
only what the protected party elects
152
Witnesses
714
712
Witnessed append-only log
154
Investigator / decision-maker system
no access to 144, 146, 708
710
Decision-maker inbox

FIG. 7

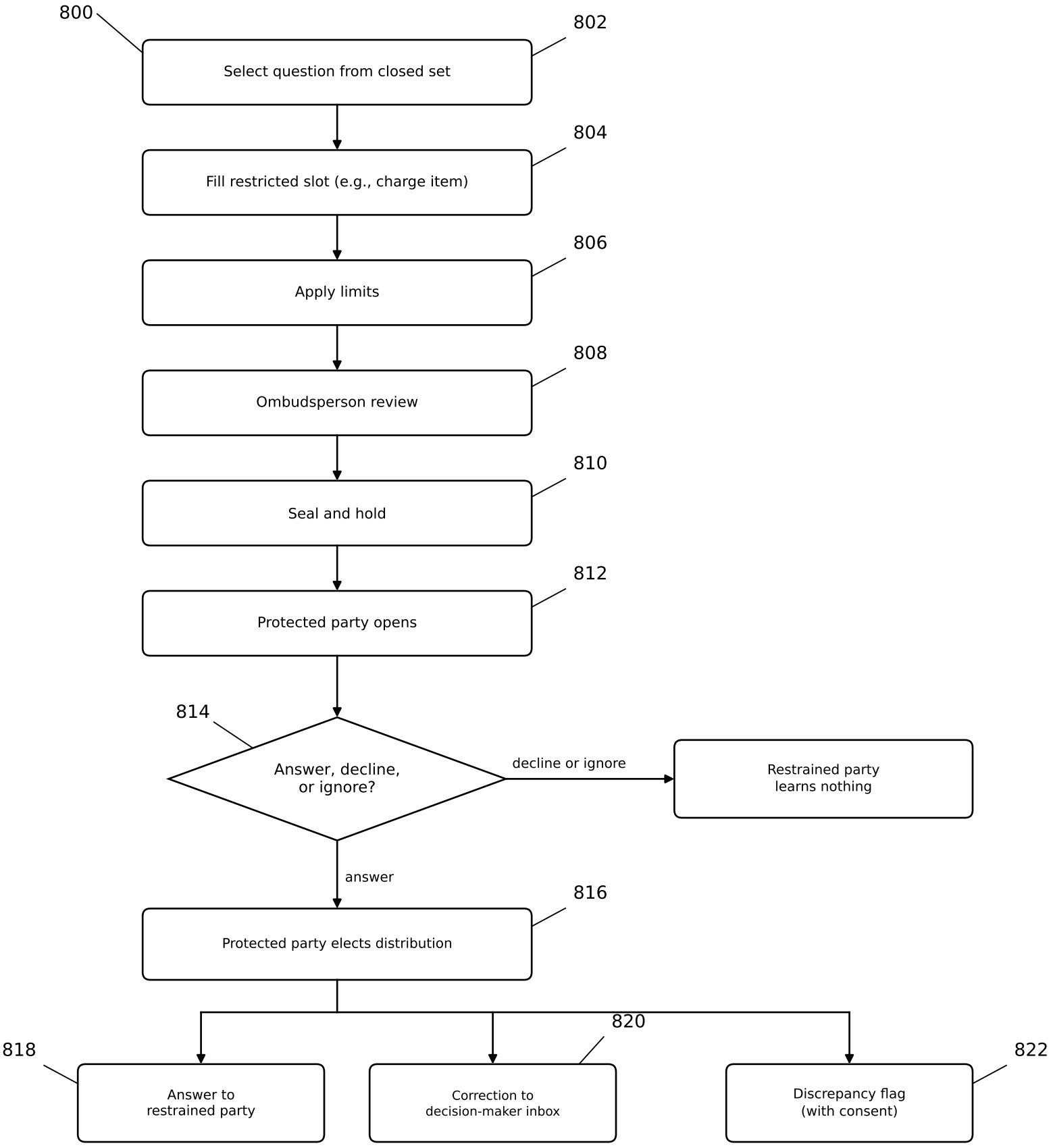

800
802
Select question from closed set
804
Fill restricted slot (e.g., charge item)
806
Apply limits
808
Ombudsperson review
810
Seal and hold
812
Protected party opens
814
Answer, decline, or ignore?
decline or ignore
Restrained party learns nothing
answer
816
Protected party elects distribution
818
Answer to restrained party
820
Correction to decision-maker inbox
822
Discrepancy flag (with consent)

FIG. 8

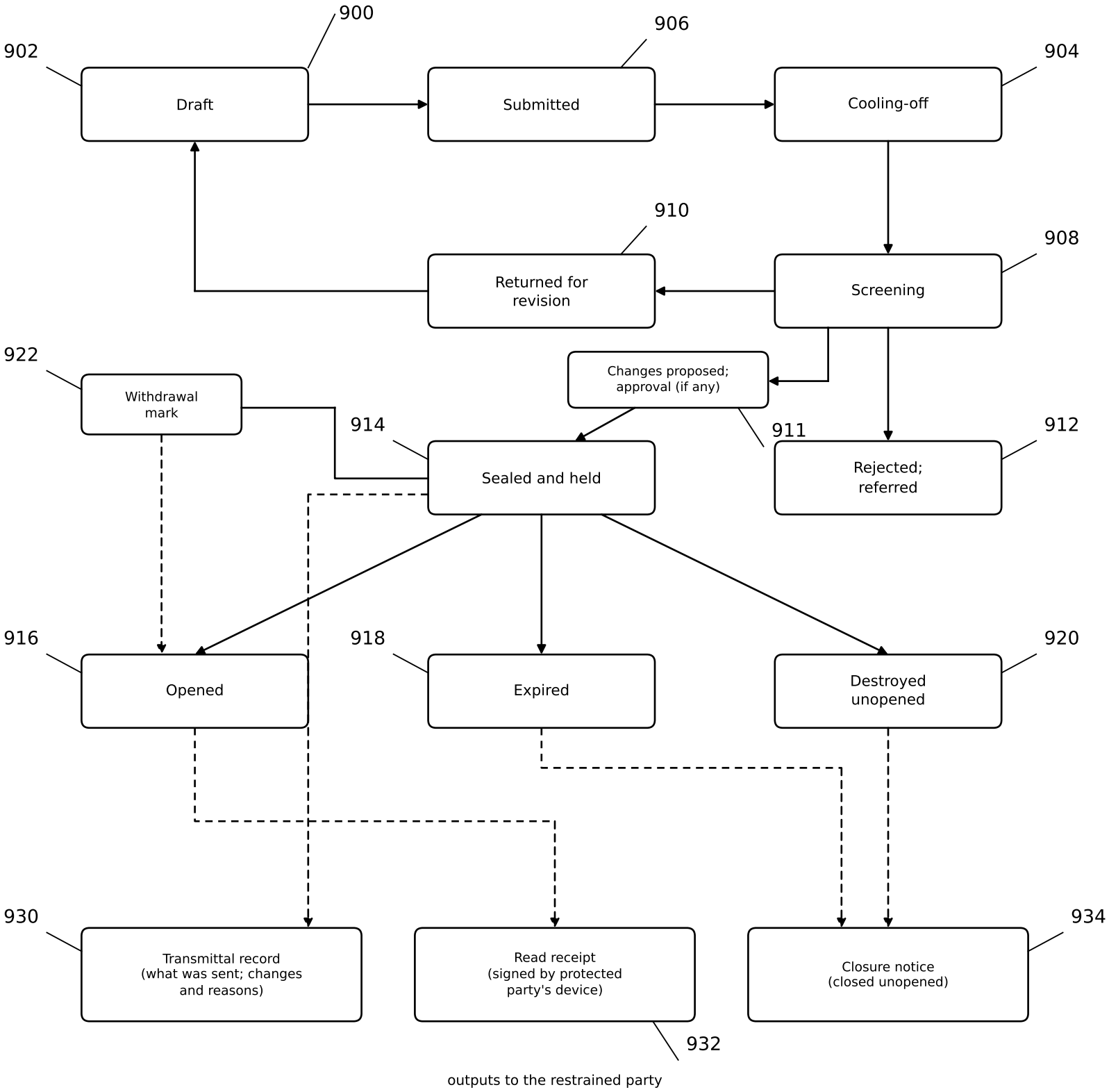

900
902
Draft
906
Submitted
904
Cooling-off
910
Returned for revision
908
Screening
922
Withdrawal mark
Changes proposed; approval (if any)
914
Sealed and held
911
912
Rejected; referred
916
Opened
918
Expired
920
Destroyed unopened
930
Transmittal record (what was sent; changes and reasons)
Read receipt (signed by protected party's device)
934
Closure notice (closed unopened)
932
outputs to the restrained party

FIG. 9

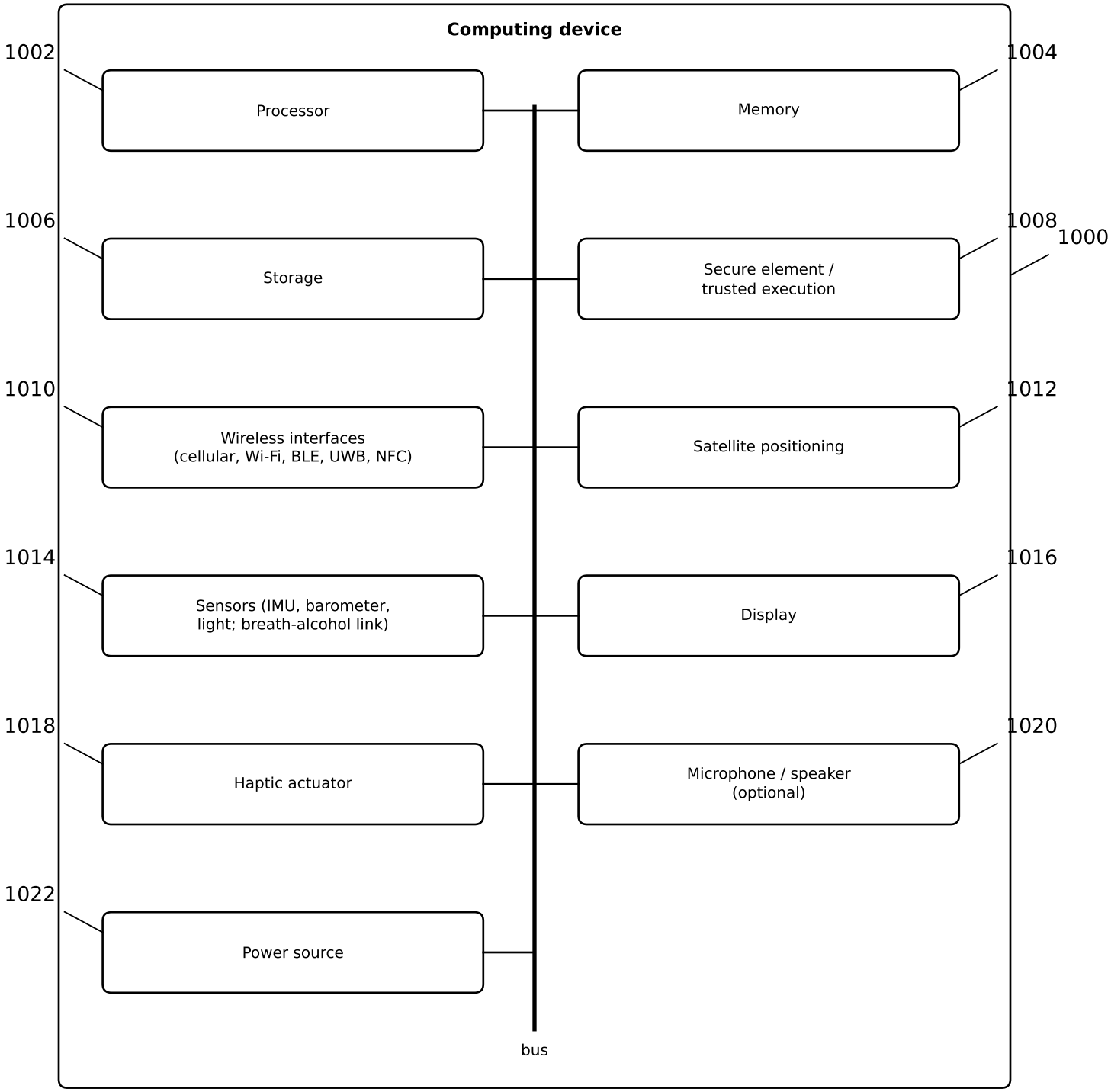

Computing device
1002
Processor
1004
Memory
1006
Storage
1008
Secure element / trusted execution
1000
1010
Wireless interfaces (cellular, Wi-Fi, BLE, UWB, NFC)
1012
Satellite positioning
1014
Sensors (IMU, barometer, light; breath-alcohol link)
1016
Display
1018
Haptic actuator
1020
Microphone / speaker (optional)
1022
Power source
bus

FIG. 10

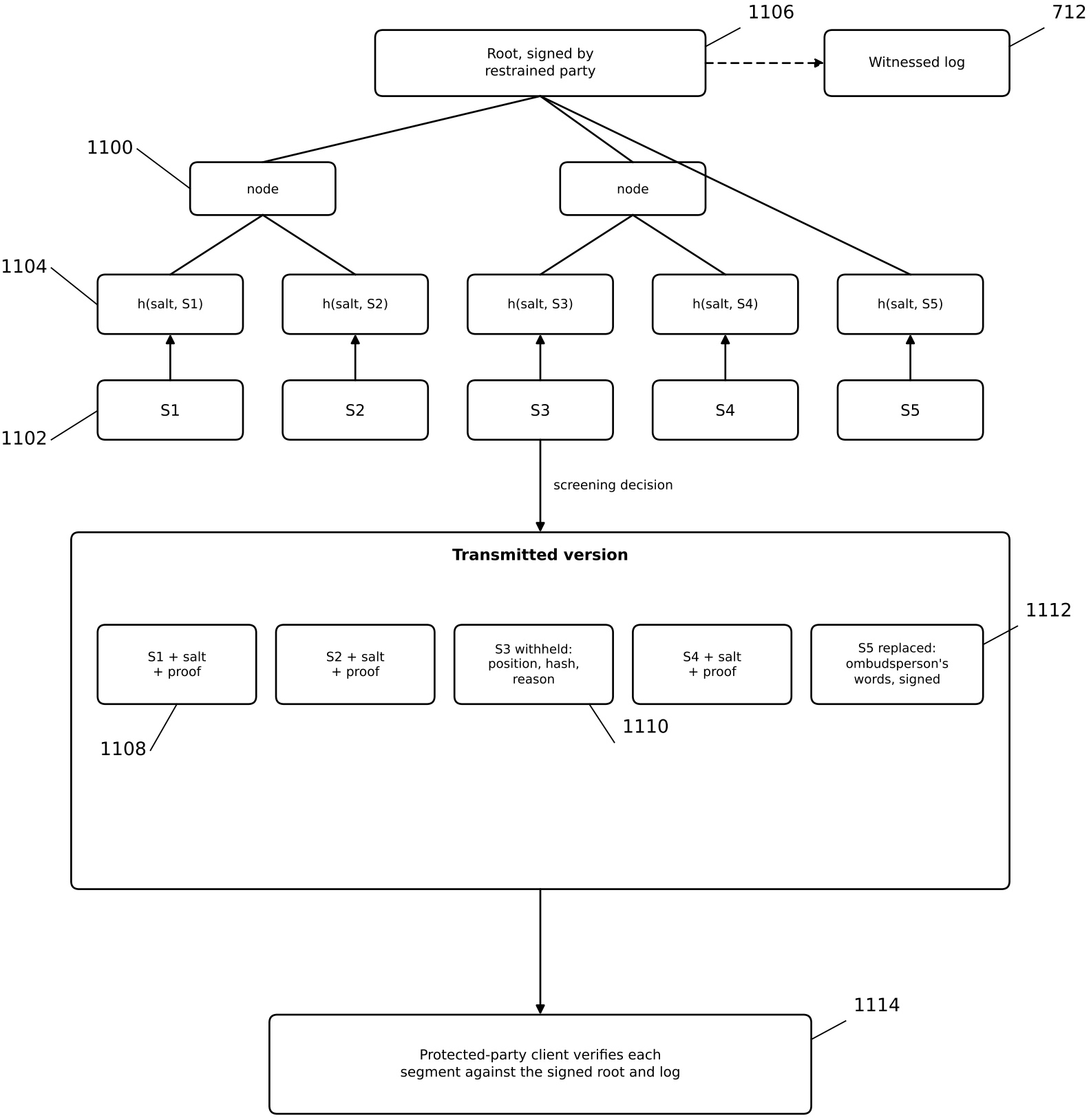
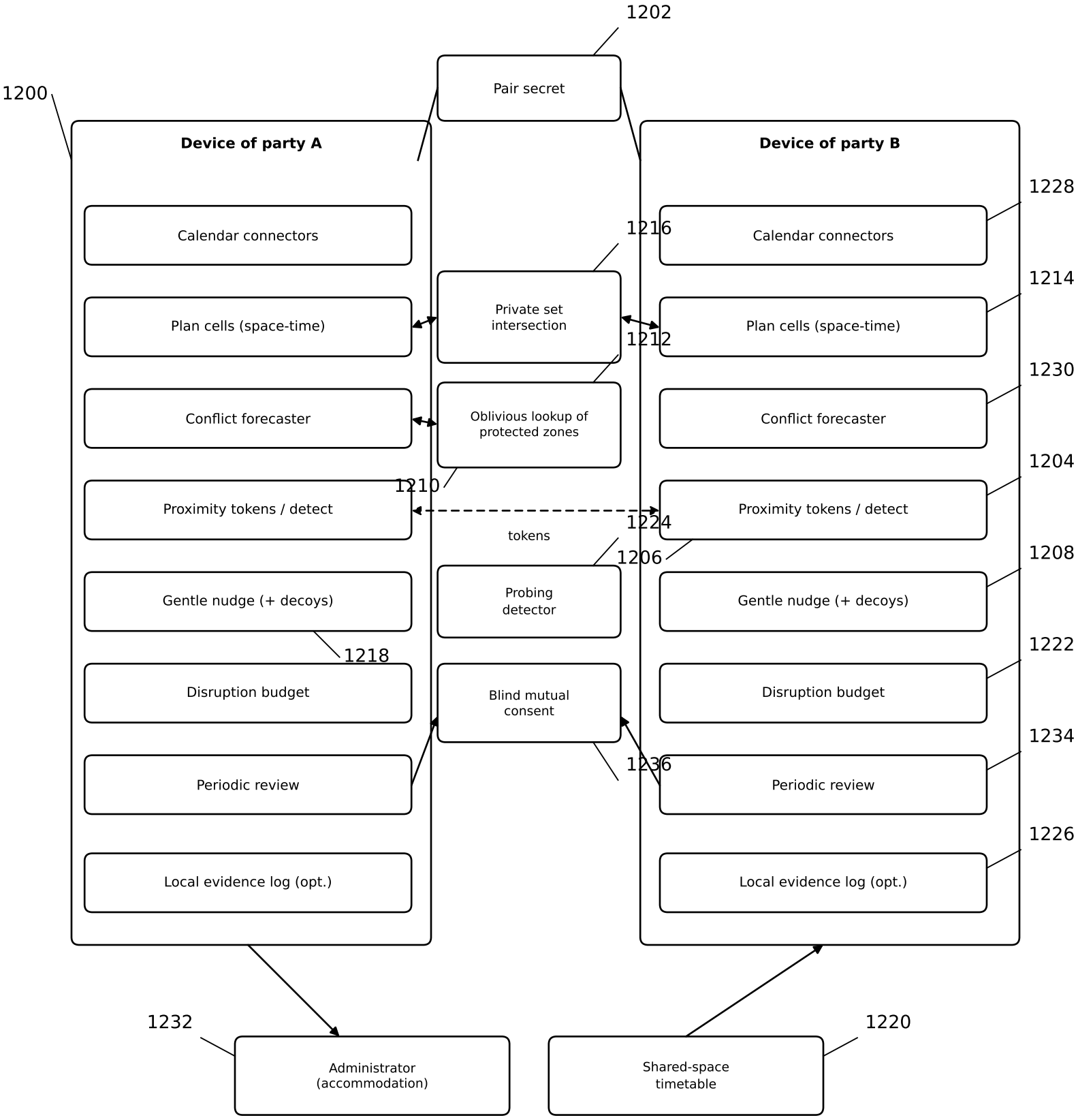

1106
712
Root, signed by restrained party
Witnessed log
1100
node
node
1104
h(salt, S1)
h(salt, S2)
h(salt, S3)
h(salt, S4)
h(salt, S5)
S1
S2
S3
S4
S5
1102
screening decision
Transmitted version
1112
S1 + salt + proof
S2 + salt + proof
S3 withheld: position, hash, reason
S4 + salt + proof
S5 replaced: ombudsperson's words, signed
1108
1110
1114
Protected-party client verifies each segment against the signed root and log

FIG. 11



no location sent to a server, the intermediary, or the other party

FIG. 12

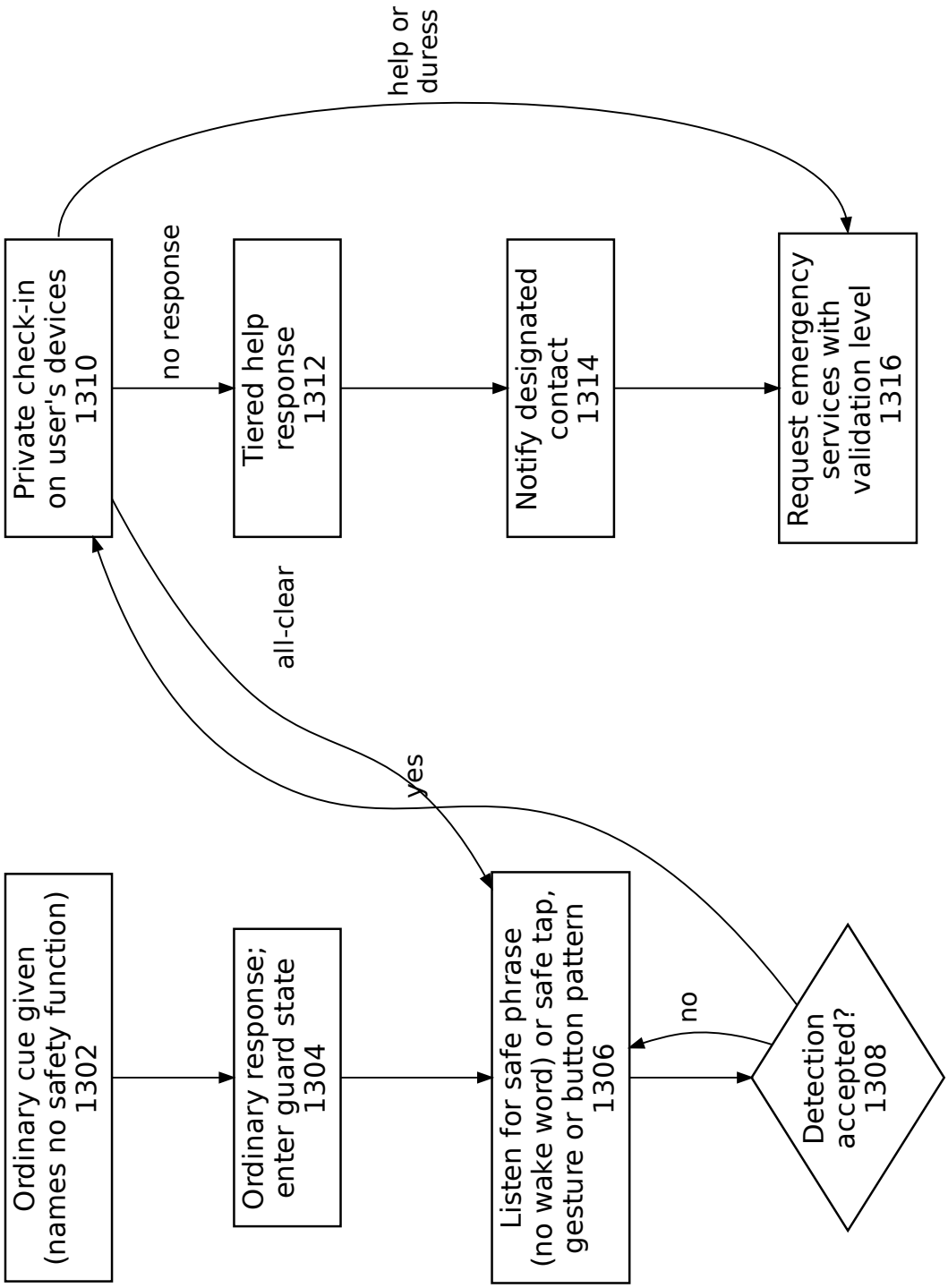

Ordinary cue given (names no safety function) 1302
Ordinary response; enter guard state 1304
Listen for safe phrase (no wake word) or safe tap, gesture or button pattern 1306
Detection accepted? 1308
no
yes
Private check-in on user's devices 1310
all-clear
no response
Tiered help response 1312
Notify designated contact 1314
help or duress
Request emergency services with validation level 1316

FIG. 13